train input, full runs — small-2way

L1D 32 sets x 2, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.33%	51.32%	55.49%	21.43%	51.50%	55.81%
parest	1/1	0.18%	0.45%	1.28%	0.49%	0.93%	7.38%
deepsjeng	1/1	2.99%	9.41%	15.98%	8.19%	23.80%	48.29%
leela	1/1	0.43%	1.55%	2.04%	6.08%	15.99%	46.72%
namd	1/1	3.22%	8.11%	17.74%	3.23%	8.13%	17.81%
lbm	1/1	9.69%	27.97%	35.99%	9.69%	27.97%	35.99%
nab	1/1	7.61%	25.91%	37.53%	8.29%	27.38%	44.95%
nab_gcn4dna	1/1	7.30%	24.89%	35.20%	8.25%	26.98%	45.65%
xz	1/1	1.36%	4.25%	4.82%	6.14%	16.52%	26.27%
xz_combined	1/1	2.79%	10.48%	25.95%	4.09%	14.43%	41.80%
imagick	1/1	0.04%	0.21%	4.19%	0.25%	0.82%	48.78%
blender	1/1	0.31%	0.55%	4.03%	3.18%	5.93%	39.04%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.21%	55.40%
omnetpp	1/1	0.01%	0.02%	0.03%	12.59%	24.64%	48.38%
xalancbmk	1/1	0.44%	1.00%	1.04%	18.29%	32.42%	63.05%
x264	1/1	6.30%	13.42%	40.78%	8.75%	18.41%	57.31%
gcc_200	1/1	1.16%	2.64%	4.34%	16.68%	35.31%	67.49%
gcc_scilab	1/1	1.25%	2.90%	4.70%	15.74%	33.65%	65.08%
gcc	1/1	1.41%	3.09%	5.21%	16.29%	32.84%	65.64%
mcf	1/1	1.61%	2.27%	13.28%	2.75%	4.04%	21.88%
perlbench_scrabbl	1/1	0.23%	0.30%	0.93%	17.65%	32.90%	56.14%
perlbench_suns	1/1	0.98%	2.06%	2.86%	18.85%	36.45%	59.85%
perlbench_diffmail	1/1	2.06%	4.51%	4.76%	17.54%	31.15%	53.97%
perlbench_perfect	1/1	1.44%	2.34%	5.00%	18.85%	33.31%	60.43%
perlbench_splitmail	1/1	5.79%	9.36%	20.68%	17.56%	30.55%	58.86%
bwaves_1	1/1	2.99%	7.65%	18.86%	2.99%	7.65%	18.86%
bwaves_2	1/1	3.04%	7.82%	18.83%	3.04%	7.82%	18.83%
exchange2	1/1	4.84%	10.67%	12.41%	6.03%	13.09%	15.72%
fotonik3d	1/1	3.99%	10.66%	12.65%	4.00%	10.69%	12.75%
roms	0/1	—	—	—	—	—	—
wrf	1/1	0.92%	3.44%	2.61%	1.59%	4.97%	8.09%
cam4	1/1	2.49%	9.15%	7.96%	4.59%	14.65%	19.40%
pooled		3.60%	9.44%	17.30%	7.34%	17.15%	41.91%
mean		3.18%	8.35%	13.49%	9.59%	20.87%	41.47%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.77%	4.46%	35.81%	3.90%	14.55%	64.32%	13.50%	8.93%
parest	done	11.51%	4.71%	12.25%	2.95%	2.54%	61.90%	1.42%	7.39%
deepsjeng	done	9.80%	2.94%	12.68%	7.87%	4.93%	6.58%	1.66%	12.65%

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
leela	done	6.41%	7.02%	7.20%	0.20%	4.07%	16.65%	0.00%	85.10%
namd	done	12.79%	5.33%	16.02%	1.39%	2.74%	25.42%	4.25%	16.19%
lbm	done	23.36%	0.42%	8.67%	2.11%	79.55%	88.88%	0.00%	7.03%
nab	done	7.24%	1.00%	11.16%	0.81%	0.72%	4.20%	0.07%	0.42%
nab_gcn4dna	done	6.93%	1.14%	10.44%	1.38%	0.69%	41.67%	1.38%	0.64%
xz	done	7.11%	1.41%	9.74%	9.93%	2.79%	32.18%	0.00%	0.93%
xz_combined	done	10.85%	3.70%	13.43%	2.30%	5.39%	20.84%	0.02%	0.77%
imagick	done	5.02%	34.99%	5.00%	0.47%	1.97%	94.86%	1.01%	4.96%
blender	done	7.42%	3.84%	8.01%	7.01%	3.57%	11.64%	2.32%	6.49%
povray	done	10.10%	7.63%	12.08%	20.37%	5.06%	41.87%	62.02%	68.89%
omnetpp	done	16.55%	0.25%	22.02%	0.87%	5.81%	28.83%	20.84%	2.89%
xalancbmk	done	12.40%	20.48%	16.05%	29.68%	4.86%	12.36%	0.00%	0.04%
x264	done	2.33%	0.39%	2.83%	0.94%	2.27%	28.09%	1.59%	13.97%
gcc_200	done	9.56%	2.26%	12.77%	5.11%	3.83%	36.44%	2.29%	7.98%
gcc_scilab	done	10.17%	2.62%	13.60%	4.68%	4.02%	36.72%	1.61%	8.78%
gcc	done	9.94%	2.41%	13.21%	4.85%	3.93%	38.95%	1.98%	8.43%
mcf	done	23.82%	7.04%	27.65%	8.49%	7.41%	75.08%	4.48%	16.24%
perlbench_scrabbl	done	7.97%	3.14%	10.83%	11.63%	3.08%	11.62%	0.00%	0.01%
perlbench_suns	done	6.55%	1.11%	9.10%	9.00%	2.40%	18.32%	0.09%	12.14%
perlbench_diffmail	done	6.46%	4.87%	8.65%	4.17%	2.66%	32.33%	0.01%	14.46%
perlbench_perfect	done	9.27%	1.24%	12.65%	5.86%	3.58%	51.16%	21.32%	24.44%
perlbench_splitmail	done	5.38%	1.79%	7.57%	2.78%	2.31%	17.25%	0.04%	3.12%
bwaves_1	done	42.24%	3.28%	46.73%	0.27%	40.60%	27.05%	5.37%	0.17%
bwaves_2	done	42.11%	3.21%	46.64%	0.27%	41.05%	27.48%	5.43%	0.17%
exchange2	done	2.26%	2.05%	2.85%	2.70%	1.43%	1.44%	0.18%	0.74%
fotonik3d	done	17.95%	5.70%	24.13%	0.06%	1.99%	38.50%	1.27%	2.89%
roms	running	—	—	—	—	—	—	—	—
wrf	done	17.59%	9.17%	18.66%	6.10%	15.15%	22.33%	6.47%	9.56%
cam4	done	11.20%	2.59%	11.85%	2.62%	12.38%	54.29%	7.88%	25.84%
pooled		11.76%	3.44%	14.32%	3.36%	7.59%	36.78%	8.06%	9.01%
mean		12.52%	4.91%	15.17%	5.19%	9.27%	34.49%	5.44%	12.01%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?

column	formula	aciklama
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.